■ Healthcare AI/ML Engineer Roadmap

Action Kit

Portfolio templates, interview questions, resume formulas, and your 90-day execution plan.

What is inside this kit:

1. 90-Day Execution Plan with weekly milestones
2. Portfolio Project Templates with starter prompts
3. Resume Bullet Formulas that translate your clinical experience
4. Interview Preparation Questions specific to this role
5. Resource Checklist with direct links

Difficulty: Very High | Timeline: 12 to 18 months | Last Updated: 2026-03-24

Built by a pharmacist who made the switch. roadmaps.tarvra.com

1. Your 90-Day Execution Plan

This plan breaks the learning path into weekly milestones. Adjust the pace to fit your schedule, but try to maintain momentum. Consistency matters more than speed.

Phase 1: Foundation

Duration: Months 1 to 4 **Effort:** 200 to 250

Week	Focus Area	Done?
Week 1	Month 1 to 2: Python Fundamentals (syntax, data types, functions, OOP), panda...	■
Week 2	Basic SQL for querying relational databases, Month 2 to 3: Statistics for Dat...	■
Week 3	Correlation vs. causation (critical in healthcare), Bias and variance, overfi...	■
Week 4	Month 3 to 4: Machine Learning Fundamentals, Supervised vs. unsupervised lear...	■

✓ **Phase Checkpoint:** Write a Python script that reads a CSV, calculates summary statistics, and filters data using a public healthcare dataset. Build a complete ML pipeline: load, split, train multiple models, evaluate, explain.

Phase 2: Technical Depth

Duration: Months 5 to 9 **Effort:** 300 to 350

Week	Focus Area	Done?
Week 5	Month 5 to 6: Deep Learning and Neural Networks, Neural network architecture,...	■
Week 6	Transformer architectures (important for medical imaging and clinical NLP), M...	■
Week 7	FHIR and HL7 standards, EHR data structure and how it differs from clean data...	■

Week 8	Clinical NLP: extracting entities from clinical notes, Month 8 to 9: ML Engin...	■

✓ **Phase Checkpoint:** Train a CNN on a medical imaging dataset (CheXpert). Parse FHIR data and extract patient information, OR build an NLP model that extracts medications from clinical notes. Add version control, parameter tracking, evaluation metrics to your deep learning model using MLflow.

Phase 3: Specialization

Duration: Months 10 to 16 **Effort:** 200 to 300

Week	Focus Area	Done?
Week 9	Track A: Medical Imaging AI (CNNs, object detection, segmentation; datasets: ...	■
Week 10	Track B: Clinical NLP and EHR Data (NER, clinical coding, cohort identificati...	■
Week 11	Track C: Predictive Analytics and Clinical Risk Models (time-series, survival...	■
Week 12	Track D: Drug Discovery and Pharmaceutical AI (molecular prediction, pharmaco...	■

✓ **Phase Checkpoint:** Polished end-to-end project in your chosen specialization, documented on GitHub.

2. Portfolio Project Templates

Each template gives you the project brief, tools needed, and starter prompts to begin immediately. Your clinical background is your competitive advantage.

Project 1: Hospital Readmission Prediction

Time Estimate: 20 to 30 hours

Build complete ML pipeline predicting 30-day readmission from patient data, including feature engineering, model selection, evaluation, and clinical validation.

Tools: Python, pandas, scikit-learn

Dataset: UCI Diabetes Readmission

✓ **Your Clinical Advantage:** You recognize that missing data patterns have clinical meaning and validate predictions against clinical intuition

Starter Prompts (begin with these questions):

- What specific problem does this project solve for clinicians or patients?
- What data would I need, and what are the realistic constraints on getting it?
- How would I measure whether my solution actually works in a clinical setting?
- What would a hiring manager want to see in my write-up of this project?

Project 2: Medical Imaging Classification

Time Estimate: 30 to 40 hours

CNN classifying chest X-rays to detect pneumonia with documented model architecture, training process, evaluation metrics, and clinical interpretation.

Tools: Python, TensorFlow, PyTorch

Dataset: CheXpert

✓ **Your Clinical Advantage:** You understand why model interpretability matters for radiologists and limitations of trained models

Starter Prompts (begin with these questions):

- What specific problem does this project solve for clinicians or patients?
- What data would I need, and what are the realistic constraints on getting it?
- How would I measure whether my solution actually works in a clinical setting?
- What would a hiring manager want to see in my write-up of this project?

Project 3: Clinical NLP: Medication Extraction

Time Estimate: 35 to 45 hours

NLP model extracting medication names and dosages from clinical notes with entity recognition, evaluated on real clinical text.

Tools: spaCy, scispaCy, Hugging Face transformers

Dataset: MIMIC-III

✓ **Your Clinical Advantage:** You recognize abbreviations and understand why context matters in clinical note analysis

Starter Prompts (begin with these questions):

- What specific problem does this project solve for clinicians or patients?
- What data would I need, and what are the realistic constraints on getting it?
- How would I measure whether my solution actually works in a clinical setting?
- What would a hiring manager want to see in my write-up of this project?

Project 4: Time-Series Patient Deterioration Prediction

Time Estimate: 40 to 50 hours

LSTM or transformer model predicting sepsis or acute decompensation from continuous vital signs with temporal pattern analysis.

Tools: PyTorch, TensorFlow, LSTM or attention models

Dataset: MIMIC time-series data or PhysioNet SEPSIS dataset

✓ **Your Clinical Advantage:** You understand clinical deterioration patterns and the value of early warning systems

Starter Prompts (begin with these questions):

- What specific problem does this project solve for clinicians or patients?
- What data would I need, and what are the realistic constraints on getting it?
- How would I measure whether my solution actually works in a clinical setting?
- What would a hiring manager want to see in my write-up of this project?

Project 5: End-to-End Specialization Project

Time Estimate: 50 to 70 hours

Polished project in your chosen track (imaging, NLP, predictive analytics, or drug discovery) with fairness analysis, model explainability, and clinical deployment considerations.

Tools: Full ML engineering stack appropriate to specialization

✓ **Your Clinical Advantage:** Deep domain expertise guides every design decision from problem framing to model validation

Starter Prompts (begin with these questions):

- What specific problem does this project solve for clinicians or patients?
- What data would I need, and what are the realistic constraints on getting it?
- How would I measure whether my solution actually works in a clinical setting?
- What would a hiring manager want to see in my write-up of this project?

3. Resume Bullet Formulas

Use these formulas to translate your clinical experience into language hiring managers recognize. Each formula follows the pattern: Clinical Skill translated to Tech Equivalent with context.

For Nurses

Clinical: Pattern recognition and clinical reasoning	**Translates to:** Model validation and error analysis: You integrate vital signs, patient appearance, context into assessment. In ML, you instinctively ask whether the model's error profile differs across patient subgroups.

How to frame it: Your clinical pattern recognition translates to model validation thinking

Clinical: HIPAA and privacy compliance	**Translates to:** Data governance and de-identification workflows: You already navigate patient privacy daily and understand why removing a name is not de-identification.

How to frame it: Your compliance knowledge applies directly to regulated ML

Clinical: Documentation under pressure	**Translates to:** Model documentation and explainability: Writing clear clinical notes translates to writing model cards and explaining outputs to non-technical clinicians.

How to frame it: Your communication skills apply to model documentation

Clinical: Workflow optimization in resource-constrained settings	**Translates to:** Pragmatic engineering and MVP thinking: Healthcare AI is full of 95% accurate models that never get used because they require data that does not exist in that hospital's workflow.

How to frame it: You understand practical constraints on AI deployment

Clinical: Medication and treatment knowledge	**Translates to:** Pharmaceutical and therapeutic AI: Deep knowledge of drug interactions and dosing protocols translates to drug-drug interaction prediction or pharmacogenomics.

How to frame it: Your pharmaceutical knowledge guides AI development

For Pharmacists

Clinical: Medication data analysis and safety	**Translates to:** Drug-drug interaction prediction and adverse event detection: Your mental checklist of contraindications and interactions becomes the foundation for building models that catch medication errors.

How to frame it: Your interaction knowledge becomes model logic

Clinical: Pharmacokinetics and pharmacodynamics	**Translates to:** Personalized medicine and precision dosing models: Understanding how drug concentrations change based on individual physiology translates to precision dosing models in oncology and critical care.

How to frame it: Your PK/PD knowledge guides precision medicine AI

Clinical: Regulatory knowledge and FDA compliance	**Translates to:** Regulated ML and FDA Software as a Medical Device (SaMD): FDA's 2021 guidance on AI/ML in software as medical device is where your regulatory literacy becomes a direct career advantage.

How to frame it: Your regulatory expertise applies to medical AI

Clinical: Clinical literature evaluation	**Translates to:** Domain-specific AI research translation: You already synthesize evidence and evaluate study design limitations.

How to frame it: Your research evaluation skills apply to AI research

Clinical: Patient education and adherence counseling	**Translates to:** Patient-facing AI and behavior change technology: You understand why generic reminders fail and what actually changes behavior.

How to frame it: You understand behavior change mechanisms

For Other Clinicians

Clinical: Complex diagnostic reasoning under uncertainty	**Translates to:** Medical imaging AI and diagnostic decision support: Understanding what 'normal variation' looks like and why a model trained on high-quality radiology scans fails in rural clinics with older equipment.

How to frame it: Your diagnostic reasoning applies to medical imaging AI

Clinical: Patient risk stratification	**Translates to:** Predictive analytics and clinical risk models: You already think about risk factors, protective factors, and how they interact.

How to frame it: Your risk thinking scales to predictive models

Clinical: Longitudinal patient care	**Translates to:** Time-series ML and temporal pattern recognition: Your intuition for how conditions evolve over time is a superpower for sequential patient data models.

How to frame it: Your longitudinal thinking applies to time-series models

Clinical: Interprofessional collaboration	**Translates to:** Healthcare AI implementation and change management: The graveyard of healthcare AI is full of technically perfect models that never got adopted.

How to frame it: You understand implementation challenges

Clinical: Documentation and evidence-based practice	**Translates to:** Clinical NLP and EHR data quality: You understand clinical language, abbreviations, and context.

How to frame it: Your language understanding helps with clinical NLP

Resume Bullet Template:
[Action verb] + [what you did in clinical terms] + [translated to tech impact] + [quantified result if possible]

Example: "Designed and maintained medication safety dashboards tracking 12,000+ monthly dispensing events, reducing near-miss medication errors by identifying data anomalies in real-time reporting workflows."

4. Interview Preparation Questions

Technical Questions

Q1: Walk me through how you would build a model to predict 30-day hospital readmission. What features would you use?

Your answer notes:

Q2: How do you handle class imbalance in a healthcare dataset where only 5% of patients have the condition?

Your answer notes:

Q3: Explain the difference between precision and recall. In a sepsis prediction model, which matters more and why?

Your answer notes:

Q4: How would you detect and mitigate bias in a clinical prediction model?

Your answer notes:

Q5: Describe your approach to model explainability for a tool that clinicians will use at the bedside.

Your answer notes:

Behavioral Questions

Q1: Tell me about a time your clinical background helped you catch a data quality issue that a pure data scientist would have missed.

Your answer notes:

Q2: How do you communicate model limitations to clinicians who expect the model to be 100% accurate?

Your answer notes:

Q3: Describe your experience working with messy EHR data. What surprised you most?

Your answer notes:

Q4: How do you stay current with both clinical practice changes and ML research?

Your answer notes:

Scenario Questions

Q1: Your model achieves 95% accuracy on the test set but performs poorly at a new hospital. What happened and how do you fix it?

Your answer notes:

Q2: A clinician says your model is wrong about a specific patient. How do you investigate?

Your answer notes:

Q3: You need to deploy a model that uses HIPAA-protected data. Walk me through your compliance approach.

Your answer notes:

5. Resource Checklist

Every resource mentioned in the roadmap, organized by learning phase. Check them off as you complete each one.

Phase 1: Foundation

Done ?	Resource	Type	Link
■	Python for Everybody	FREE	https://www.py4e.com/
■	Kaggle Python course	FREE	https://www.kaggle.com/learn/python
■	SQLZoo	FREE	https://sqlzoo.net/
■	StatQuest with Josh Starmer	FREE	https://www.youtube.com/c/joshstarmer
■	Seeing Theory	FREE	http://students.brown.edu/seeing-theory/
■	Google ML Crash Course	FREE	https://developers.google.com/machine-learnin...
■	Fast.ai Practical Deep Learning	FREE	https://course.fast.ai/
■	scikit-learn tutorials	FREE	https://scikit-learn.org/stable/getting_start...
■	Andrew Ng ML Specialization on Coursera	PAID	https://www.coursera.org/specializations/mach...

Phase 2: Technical Depth

Done ?	Resource	Type	Link
■	3Blue1Brown Neural Networks series	FREE	https://www.youtube.com/channel/UCYO_jab_esuF...
■	Fast.ai Part 2	FREE	https://course.fast.ai/
■	TensorFlow tutorials	FREE	https://www.tensorflow.org/tutorials
■	DeepLearning.AI Specialization on Coursera	PAID	https://www.deeplearning.ai/
■	spaCy NLP tutorials	FREE	https://spacy.io/usage
■	scispaCy for medical NLP	FREE	https://allenai.github.io/scispacy/
■	ClinicalBERT	FREE	https://github.com/kexinhuang12345/ClinicalBERT

■	Made With ML	FREE	https://madewithml.com/
■	MLflow documentation	FREE	https://mlflow.org/docs/latest/index.html

Phase 3: Specialization

Done ?	Resource	Type	Link
■	Medical Imaging datasets and resources	FREE	https://stanfordmlgroup.github.io/competition...
■	MIMIC-III and MIMIC-IV datasets	FREE	https://mimic.physionet.org/
■	DeepChem	FREE	https://deepchem.io/
■	RDKit	FREE	https://www.rdkit.org/

What to do next:

1. Take the Career Quiz to confirm this role fits you: quiz.ehoneahobed.com
2. Subscribe to The Transmutation for weekly insights: thetransmutation.substack.com
3. Complete Move 1 from your roadmap today. It takes less than 3 hours.